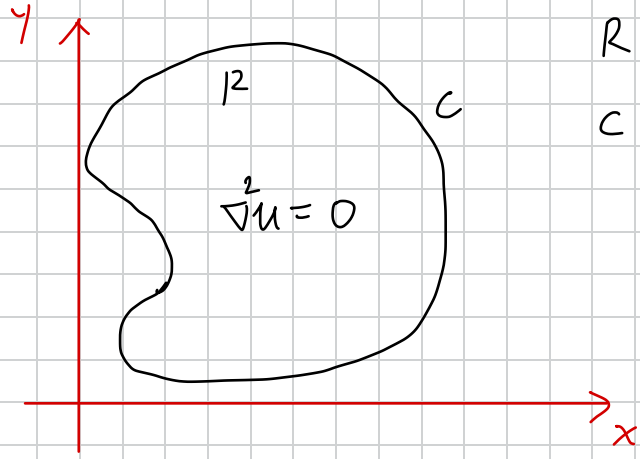
Solución numérica de EDP

Ecuación de Laplace: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

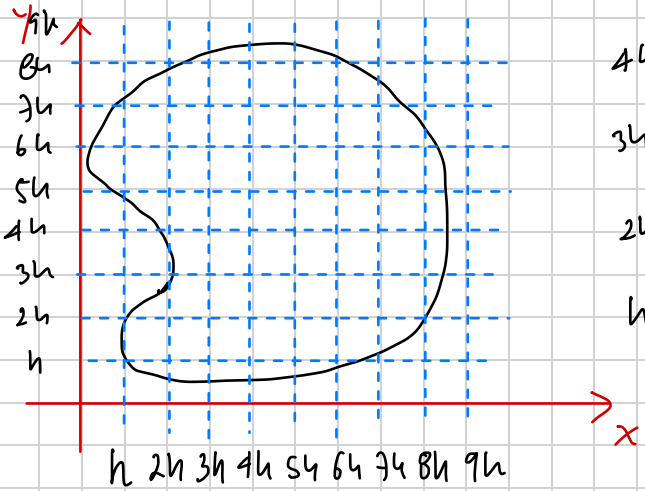
Ecuación de calor $K \frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$; $K > 0$

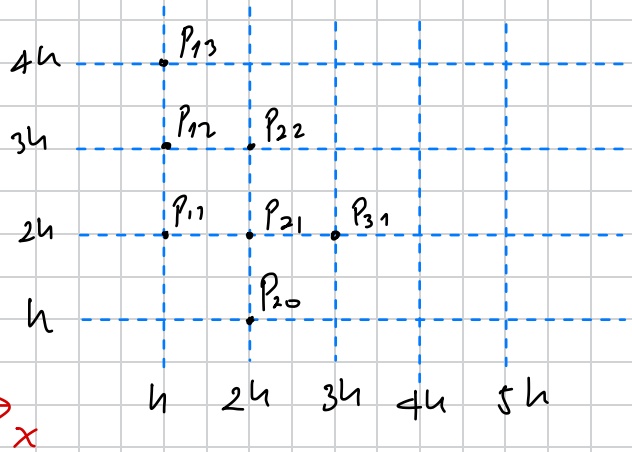
Ecuación de onda: $a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$

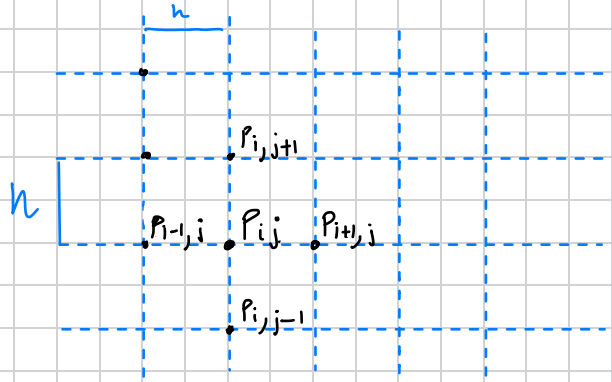


R la región (Ω)

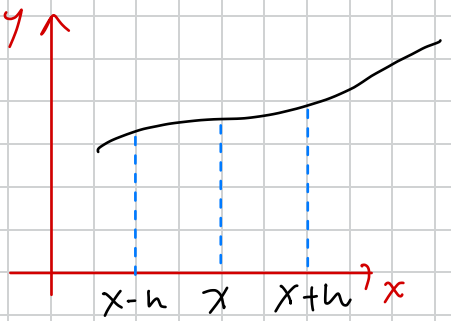
C es la frontera $(\partial\Omega)$







Sea $\nabla^2 u = 0$ y sean las aproximaciones de las segundas derivadas parciales usando diferencias divididas centrales:



$f(x)$

$f(x+h)$

$$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$$

$$f'(x) = \frac{f(x+h)-f(x)}{h}$$

$$f'(x) = \frac{f(x)-f(x-h)}{h}$$

$$f'(x) = \frac{f(x+h)-f(x-h)}{h}$$

$$f''(x) = \frac{f(x+h)-2f(x)+f(x-h)}{h^2}$$

Diferencias divididas centrales.

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{h^2}\left(u(x+h,y)-2u(x,y)+u(x-h,y)\right)$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{1}{h^2}\left(u(x,y+h)-2u(x,y)+u(x,y-h)\right)$$

$$\nabla^2 u = 0 \quad ; \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{1}{h^2}\left(u(x+h,y)-2u(x,y)+u(x-h,y)\right) + \frac{1}{h^2}\left(u(x,y+h)-2u(x,y)+u(x,y-h)\right) = 0$$

$$\frac{1}{h^2}\left(u(x+h,y)-4u(x,y)+u(x-h,y)+u(x,y+h)+u(x,y-h)\right) = 0$$

$$u(x+h,y)+u(x-h,y)+u(x,y+h)+u(x,y-h)-4u(x,y) = 0 \quad (1)$$

Vamos a usar la siguiente notación:

$u(x,y) = u_{ij}$; $u(x+h,y) = u_{i+1,j}$; $u(x-h,y) = u_{i-1,j}$

$u(x,y+h) = u_{i,j+1}$; $u(x,y-h) = u_{i,j-1}$

$P_{ij} = P(ih, jh)$ i contador entero en el eje x

$0, 1, 2, \ldots, n$

j contador entero en el eje y.

$0, 1, 2, \ldots, n$

$P_{20} = P(2h, 0)$; $P_{11} = P(h, h)$; $P_{21} = P(2h, h)$;

$P_{22} = P(2h, 2h)$.

Así, la ecuación (1) se reescribe:

$$u(x+h, y) + u(x-h, y) + u(x, y+h) + u(x, y-h) - 4u(x, y) = 0 \quad (1)$$

$$u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1} - 4u_{ij} = 0 \quad (2)$$

De la ecuación (2) queremos despejar u_{ij}

$$u_{ij} = \frac{1}{4}\left(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}\right) \quad (3)$$

La ecuación (3) es la ecuación de recurrencia para resolver $\nabla^2 u = 0$.

Problema Dirichlet: Se conocen los valores de u en la frontera. Esto reduce la cantidad de valores u_{ij} a calcular.

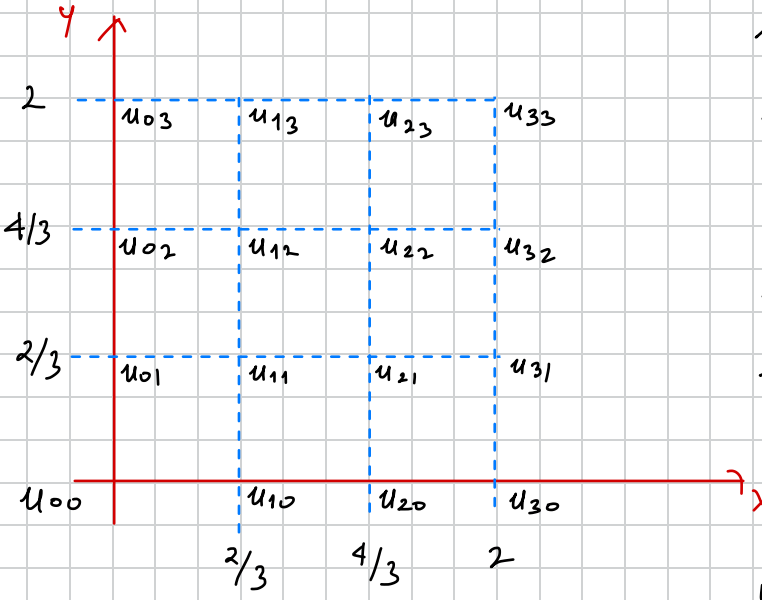
Ejemplo: Resolver EDP:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad ; \quad 0 < x < 2 \; ; \quad 0 < y < 2.$$

$$u(0, y) = 0 \quad ; \quad u(2, y) = y(2 - y)$$

$$u(x, 0) = 0 \quad ; \quad u(x, 2) = \begin{cases} x & ; \quad 0 < x < 1 \\ 2 - x & ; \quad 1 < x < 2 \end{cases}$$

Considerar $h = 2/3$ (Partición uniforme).



$u_{00} = u_{10} = u_{20} = u_{30} = 0$

$u_{01} = u_{02} = u_{03} = 0$

$u(2,y) = y(2-y)$

$u_{31} = 2/3\,(2 - 2/3) = 8/9$

$u_{32} = 4/3\,(2 - 4/3) = 8/9$

$u_{33} = 2\,(2-2) = 0$

$u_{13} = u(2/3, 2) = 2/3$

$u_{23} = u(4/3, 2) = 2/3$

$$u(x,2) = \begin{cases} x; & 0 < x < 1 \\ 2-x; & 1 < x < 2 \end{cases}$$

$$u_{ij} = \frac{1}{4}\left(u_{i+1,j} + u_{i-1,j} + u_{i,j+1} + u_{i,j-1}\right)$$

$$u_{11} = \frac{1}{4}(u_{21} + u_{01} + u_{12} + u_{10}) \quad ; \quad 4u_{11} = u_{21} + u_{12}$$

$$-4u_{11} + u_{21} + u_{12} = 0 \quad (*)$$

$$u_{12} = \frac{1}{4}(u_{22} + u_{02} + u_{13} + u_{11}) \quad ; \quad 4u_{12} = u_{22} + 2/3 + u_{11}$$

$$u_{11} - 4u_{12} + u_{22} = -2/3 \quad (**)$$

$$u_{21} = \frac{1}{4}(u_{31} + u_{11} + u_{22} + u_{20}) \quad ; \quad 4u_{21} = \frac{8}{9} + u_{11} + u_{22}$$

($u_{31} = 8/9$)

$$u_{11} - 4u_{21} + u_{22} = -\frac{8}{9} \quad (***)$$

$$u_{22} = \frac{1}{4}(u_{32} + u_{12} + u_{23} + u_{21}) \quad ; \quad 4u_{22} = u_{12} + u_{21} + \frac{14}{9}$$

($u_{32} = 8/9$, $u_{23} = 2/3$)

$$-4u_{22} + u_{21} + u_{12} = -\frac{14}{9} \quad (****)$$

$$-4u_{11} + u_{21} + u_{12} = 0 \quad (*)$$

$$u_{11} - 4u_{12} + u_{22} = -2/3 \quad (**)$$

$$u_{11} - 4u_{21} + u_{22} = -\frac{8}{9} \quad (***)$$

$$-4u_{22} + u_{21} + u_{12} = -\frac{14}{9} \quad (****)$$

$$\begin{aligned} -4u_{11} + u_{21} + u_{12} &= 0 \\ u_{11} - 4u_{21} + u_{22} &= -8/9 \\ u_{11} - 4u_{12} + u_{22} &= -2/3 \\ u_{21} + u_{12} - 4u_{22} &= -14/9 \end{aligned}$$

$$\begin{bmatrix} -4 & 1 & 1 & 0 \\ 1 & -4 & 0 & 1 \\ 1 & 0 & -4 & 1 \\ 0 & 1 & 1 & -4 \end{bmatrix} \begin{bmatrix} u_{11} \\ u_{21} \\ u_{12} \\ u_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ -8/9 \\ -2/3 \\ -14/9 \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1}B.$$

$$\begin{bmatrix} u_{11} \\ u_{21} \\ u_{12} \\ u_{22} \end{bmatrix} = \begin{bmatrix} 0.194 \\ 0.417 \\ 0.361 \\ 0.583 \end{bmatrix}$$

TAREA: Solucionar EDP:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad ; \quad 0 < x < 2 \; ; \; 0 < y < 2.$$

$$u(0,y) = 0 \; ; \quad u(2,y) = y(2-y)$$

$$u(x,0) = 0 \; ; \quad u(x,2) = \begin{cases} x & ; \quad 0 < x < 1 \\ 2-x & ; \quad 1 < x < 2 \end{cases}$$

Considerar $h = 1/2$ (Partición uniforme).

Ecuación diferencial de calor

$$c\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{h^2}\left(u(x+h,t) - 2u(x,t) + u(x-h,t)\right)$$

$$\frac{\partial u}{\partial t} = \frac{1}{k}\left(u(x,t+k) - u(x,t)\right)$$

$$\frac{c}{h^2}\left(u(x+h,t) - 2u(x,t) + u(x-h,t)\right) = \frac{1}{k}\left(u(x,t+k) - u(x,t)\right)$$

$$\frac{ck}{h^2}\left(u(x+h,t) - 2u(x,t) + u(x-h,t)\right) = \left(u(x,t+k) - u(x,t)\right)$$

Vamos a usar la siguiente notación:

$u(x,t) = u_{ij}$; $u(x+h,t) = u_{i+1,j}$; $u(x-h,t) = u_{i-1,j}$

$u(x,t+k) = u_{i,j+1}$. Sea $\lambda = \frac{ck}{h^2}$

$$\lambda u_{i+1,j} - 2\lambda u_{ij} + \lambda u_{i-1,j} = u_{ij+1} - u_{ij}$$

$$u_{i,j+1} = \lambda u_{i+1,j} + (1-2\lambda)u_{ij} + \lambda u_{i-1,j}$$

Ecuación de recurrencia.

$0 \le x \le a$; $t \ge 0$ $0 \le t \le T$

$h = \frac{a}{n}$ $k = \frac{T}{m}$

$x_i = ih$ $t_j = jk$

$i = 0, 1, 2, \ldots, n$ $j = 0, 1, 2, \ldots, m$

Ejemplo: $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$; $0 < x < 1$

$0 < t < 1/2$

$u(x,0) = 0$, $u(1,t) = 0$; $0 \leq t \leq 1/2$

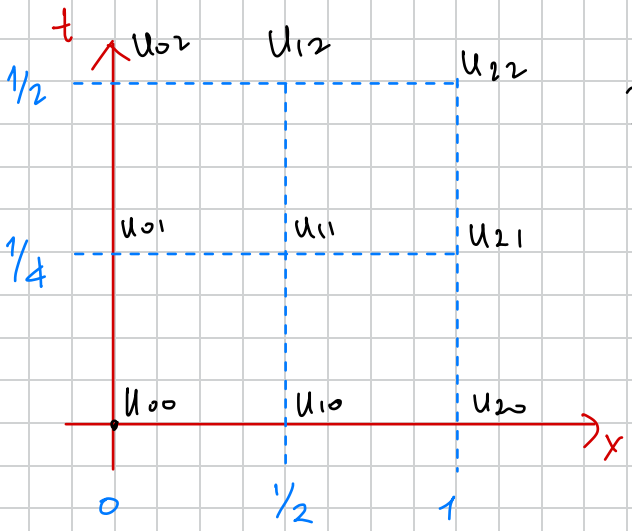
$u(x,0) = \text{Sen}(\pi x)$; $0 \leq x \leq 1$

$h = 1/2$; $h = \frac{a}{n}$; $n = \frac{a}{h}$

; $n = \frac{1}{1/2}$; $n = 2$

$k = \frac{T}{m}$;

	i					
$x =$	0	1	2	0	1/2	1
	j					
$t =$	0	1	2	0	1/4	1/2



$\lambda = \frac{ck}{h^2} = \frac{(1)(1/4)}{(1/2)^2} = 1$

$u_{i,j+1} = \lambda u_{i+1,j} + (1-2\lambda) u_{ij} + \lambda u_{i-1,j}$

$i = 1$; $j = 0$

$u_{1,1} = \lambda u_{2,0} + (1-2\lambda) u_{1,0} + \lambda u_{0,0}$

$u_{11} = (1-2\lambda) \text{Sen}(\pi/2)$; $u_{11} = -1$

$i = 1$; $j = 1$

$u_{12} = \lambda u_{21} + (1-2\lambda) u_{11} + \lambda u_{01}$

$u_{12} = (1-2\lambda) u_{11}$, $u_{12} = 1$

TAREA: $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$; $0 < x < 1$

$0 < t < 1/2$

$u(x,0) = 0$, $u(1,t) = 0$; $0 \leq t \leq 1/2$

$u(x,0) = \text{Sen}(\pi x)$; $0 \leq x \leq 1$

$m = n = 4$

Ecuación diferencial de onda

$$c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \qquad 0 \leq x \leq a \qquad 0 \leq t \leq T$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{h^2}\left(u(x+h,t) - 2u(x,t) + u(x-h,t)\right)$$

$$\frac{\partial^2 u}{\partial t^2} = \frac{1}{k^2}\left(u(x,t+k) - 2u(x,t) + u(x,t-k)\right)$$

$0 \leq x \leq a$; $t \geq 0$ $0 \leq t \leq T$ Sea $\lambda = \frac{ck}{h}$

$h = \frac{a}{n}$ $\qquad k = \frac{T}{m}$

$x_i = ih$ $\qquad t_j = jk$

$i = 1, 2, \ldots, n-1$ $\qquad j = 1, 2, \ldots, m-1$

Ecuación de recurrencia.

$$u_{i,j+1} = \lambda^2 u_{i+1,j} + 2(1-\lambda^2) u_{ij} + \lambda^2 u_{i-1,j} - u_{i,j-1}$$

$$g(x_i) = u_t(x_i, 0) \approx \frac{u(x_i, k) - u(x_i, -k)}{2k}$$

$$u(x_i, -k) = u(x_i, k) - 2kg(x_i)$$

$$u_{i,-1} = u_{i,1} - 2kg(x_i)$$

$$u_{i,1} = \frac{\lambda^2}{2}(u_{i+1,0} + u_{i-1,0}) + (1-\lambda^2) u_{i,0} + kg(x_i)$$

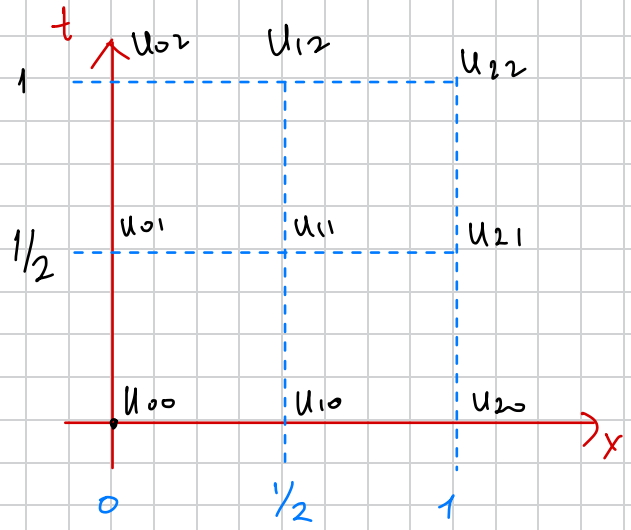
Ejemplo: $\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}$ $\quad 0 < x < 1$, $\quad 0 < t < 1$

$u(0,t) = 0$; $u(1,t) = 0$; $0 \leq t \leq 1$

$u(x,0) = \text{Sen}(\pi x)$; $\left.\frac{\partial u}{\partial t}\right|_{t=0} = 0$; $0 \leq x \leq 1$

Usar igual cantidad de divisiones.

x	$=$	0	$1/2$ $\quad$ 1
t	$=$	0	$1/2$ $\quad$ 1



$\lambda = \frac{ck}{h} = 1$

$u_{i,j+1} = \lambda^2 u_{i+1,j} + 2(1-\lambda^2) u_{ij} + \lambda^2 u_{i-1,j} - u_{i,j-1}$

$i = 1$; $j = 0$

$u_{11} = \lambda^2 u_{20} + 2(1-\lambda^2) u_{10} + \lambda^2 u_{00} - u_{1,-1}$

$u_{i,-1} = u_{i,1} - 2k\, g(x_i)$

$u_{1,-1} = u_{11} - 2k\, g(x_1)$

$u_{11} = 2(1-\lambda^2) u_{10} - u_{11} + 2k\, g(x_1)$

$u_{11} = (1-\lambda^2) u_{10} + k\, g(x_1)$

$u_{11} = (1-\lambda^2)(1) + k(1)$

$u_{11} = 1/2$

$$u_{i,j+1} = \lambda^2 u_{i+1,j} + 2(1-\lambda^2) u_{ij} + \lambda^2 u_{i-1,j} - u_{i,j-1}$$

$i=1$; $j=1$

$$u_{12} = \lambda^2 u_{21} + 2(1-\lambda^2) u_{11} + \lambda^2 u_{01} - u_{10}$$

$$u_{12} = 1. \qquad 1$$

TAREA:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} \qquad 0 < x < 1 \qquad 0 < t < 1$$

$$u(0,t) = 0 \quad ; \quad u(1,t) = 0 \; ; \quad 0 \leq t \leq 1$$

$$u(x,0) = \text{Sen}(\pi x) \quad ; \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = 0 \quad ; \quad 0 \leq x \leq 1$$

Usar igual cantidad de divisiones.

x	=	0	1/4	1/2	3/4	1
t	=	0	1/4	1/2	3/4	1.